

FIRST SEMESTER (CBCSS) DEGREE EXAMINATION, NOVEMBER 2020**Mathematics****MEC 1C 01—MATHEMATICAL ECONOMICS****(2019 Admissions)****(Multiple Choice Questions for SDE Candidates)****Time : 15 Minutes****Total No. of Questions : 15****Maximum : 15 Marks****INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 15.
2. The candidate should check that the question paper supplied to him/her contains all the 15 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MEC 1C 01--MATHEMATICAL ECONOMICS

(Multiple Choice Questions for SDE Candidates)

1. Which of the following are NOT true ?

- (A) $f(x) = ax^n$ implies $f'(x) = anx^{n-1}$.
- (B) $f(x) = 4x^5 - 3x^2$ implies $f'(x) = 20x^5 - 6x^{-2}$.
- (C) $f(x) = 4x + 3/x^2$ implies $f'(x) = 4 - 6/x^3$.
- (D) $f(x) = \ln(x)$ implies $f'(x) = x^{-1}$.

2. Which of the following statements are (in general) true ?

- (A) Marginal cost (MC) is minimised where MC = Average Variable Cost (AVC).
- (B) Total Cost (ATC) is minimised where MC = ATC.
- (C) Average Variable Cost (AVC) is minimised where MC = AVC.
- (D) Total revenue is maximised where MC = Marginal Revenue (MR).

3. The law which studies the direct relationship between price and quantity supplied of a commodity is :

- (A) Law of demand.
- (B) Law of variable proportion.
- (C) Law of supply.
- (D) None of the above.

4. In case of perfectly inelastic supply the supply curve will be :

- (A) Rising.
- (B) Vertical.
- (C) Horizontal.
- (D) Falling.

5. When a percentage in price results in equal change in quantity supplied, it is called :

- (A) Elastic supply.
- (B) Perfectly inelastic.
- (C) Elasticity of supply
- (D) Unitary elastic supply.

6. Suppose the supply for product A is perfectly elastic. If the demand for this product increases :
- (A) The equilibrium price and quantity will increase.
 - (B) The equilibrium price and quantity will decrease.
 - (C) The equilibrium quantity will increase but the price will not change.
 - (D) The equilibrium price will increase but the quantity will not change.
7. If the co-efficient of income elasticity of demand is higher than 1 and the revenue increases, the share of expenditures for commodity X in total expenditure :
- (A) Will increase.
 - (B) Will decrease.
 - (C) Will remain constant.
 - (D) Can not be determined.
8. If the demand curve for product A moves to the right, and the price of product B decreases, it can be concluded that :
- (A) A and B are substitute goods.
 - (B) A and B are complementary goods.
 - (C) A is an inferior good, and B is a superior good.
 - (D) Both goods A and B are inferior.
9. If a price increase of 50 % results in an increase in the quantity supplied of an economic good from 10 to 20 pieces, calculate the co-efficient of price elasticity of supply :
- (A) $1/4$.
 - (B) $1/2$.
 - (C) 1.
 - (D) 2.
10. Which of the following statements is false :
- (A) Perfect competition involves many sellers of standardized products.
 - (B) Monopolistic competition involves many sellers of homogeneous products.
 - (C) The oligopoly involves several producers of standardized or differentiated products.
 - (D) Monopoly involves a single product for which there are no close substitutes.
11. If the price elasticity of demand for wine is estimated to be -0.6 , then a 20 % increase in price of wine will lead to _____ in quantity demanded of wine at that price.
- (A) 12 % increase.
 - (B) 12 % decrease.
 - (C) 19.6 % increase.
 - (D) 20.6 % decrease.

12. If the cross price elasticity of demand for two product is negative, then the two products are _____.
- (A) Complementary to each other. (B) Perfectly substitute for each other.
(C) Completely competitive. (D) Unrelated.
13. When the price of complementary products falls, the demand of the other product will :
- (A) Fall. (B) Increases.
(C) Remain stable. (D) Drops by 25 %.
14. Goods which are perfect substitute of each other will have elasticity of substitution :
- (A) Unity. (B) Less than 1.
(C) More than 1. (D) Infinite.
15. In question No. 201 if at 15,000, the dealer is prepared to supply on 1250 sets of TV the elasticity of supply is :
- (A) 1. (B) 2.
(C) 0.75. (D) 1.4.

FIRST SEMESTER (CBCSS) DEGREE EXAMINATION, NOVEMBER 2020**Mathematics****MEC 1C 01—MATHEMATICAL ECONOMICS****(2019 Admissions)****Time : Two Hours****Maximum : 60 Marks****Section A***Answer at least **eight** questions.**Each question carries 3 marks.**All questions can be attended.**Overall Ceiling 24.*

1. What is a Utility function ?
2. Define Long run costs.
3. Define Law of supply.
4. Give the maning of MRS_{YX} .
5. What is an Indifference Curve ?
6. Explain Cross elasticity of demand.
7. If $TC = 8Q^2 + 10Q + 15$, what will be the MC ?
8. Define Consumer equilibrium.
9. What is meant by Investment multiplier ?
10. Explain Optimization.
11. Explain Arc elasticity method.
12. What is meant by Shift in demand ?

(8 × 3 = 24 marks)**Turn over**

Section B

Answer at least five questions.

Each question carries 5 marks.

All questions can be attended.

Overall Ceiling 25.

13. Derive the relationship between MC and AC
14. What is income elasticity of demand ? The demand for a commodity is given by $Q = 5000 + 4Y$, where Q is the demand and Y is the income. What will be the income elasticity of demand when the income of the consumer is Rs. 7000 ?
15. What is a demand curve ? Why does a demand curve slope downward ?
16. Explain long run average cost curve. Why it is known as a planning curve ?
17. State and explain the law of diminishing marginal utility. Show how law of demand can be derived from it.
18. What is a budget line ? Derive the slope of a budget line.
19. What is marginal productivity ? Using the production function $Q = AL^aK^b$, show that, if the factors of production are paid according to their marginal product, the total product will be exhausted. (Q = total product = labor, K = capital)

(5 × 5 = 25 marks)

Section C

Answer any one questions.

The question carries 11 marks.

20. Explain cardinal utility analysis of demand. Derive consumer equilibrium using cardinal utility method.
21. Explain the significance of Lagrange multiplier and maximize the function $X_1X_2 + 2X_1$ Subject to the constraint $4X_1 + 2X_2 = 60$.

(1 × 11 = 11 marks)

**FIRST SEMESTER (CBCSS—UG) DEGREE EXAMINATION
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Mathematics

MEC 1C 01—MATHEMATICAL ECONOMICS

(2019—2020 Admissions)

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes

Total No. of Questions : 15

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MEC 1C 01—MATHEMATICAL ECONOMICS

(Multiple Choice Questions for SDE Candidates)

1. Which of the following are NOT true ?
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- (D) $f(x) = \ln(x)$ implies $f'(x) = x^{-1}$.
2. The law which studies the direct relationship between price and quantity supplied of a commodity is :
- (A) Law of demand. (B) Law of variable proportion.
- (C) Law of supply. (D) None of the above.
3. In case of perfectly inelastic supply the supply curve will be :
- (A) Rising. (B) Vertical.
- (C) Horizontal. (D) Falling.
4. When a percentage in price results in equal change in quantity supplied, it is called ?
- (A) Elastic supply. (B) Perfectly inelastic.
- (C) Elasticity of supply. (D) Unitary elastic supply.
5. If the co-efficient of income elasticity of demand is higher than 1 and the revenue increases, the share of expenditures for commodity X in total expenditure :
- (A) Will increase. (B) Will decrease.
- (C) Will remain constant. (D) Can not be determined.
6. Calculate the average fixed cost (AFC), for a level of production $Q = 20$, knowing that the total cost function is : $TC = 200 + 3Q + 2Q^2$.
- (A) 1060. (B) 200.
- (C) 20. (D) 10.

7. On the market with perfect competition :
- (A) The firm is a "price-taker," meaning, it takes over the market price.
 - (B) The firm is a "price-maker", meaning, it determines the market price.
 - (C) The companies' products are differentiated.
 - (D) Input barriers are minimal, and exit barriers are maximal.
8. If the price elasticity of demand for wine is estimated to be -0.6 , then a 20 % increase in price of wine will lead to _____ in quantity demanded of wine at that price.
- (A) 12 % increase.
 - (B) 12 % decrease.
 - (C) 19.6 % increase.
 - (D) 20.6 % decrease.
9. If the price of coffee falls by 8 % and the demand for Tea declines by 2 %. The cross price elasticity of demand for Tea is :
- (A) 0.45.
 - (B) 0.25.
 - (C) + 0.44.
 - (D) -0.30 .
10. When the price of complementary products increases, the demand of the other product will ?
- (A) Falls.
 - (B) Increases.
 - (C) Remains same.
 - (D) Increases by 25 %.
11. An individual is spending his entire income on two items A and B equally. If income elasticity of A is 4 what is income elasticity of B :
- (A) 4.
 - (B) 2.
 - (C) 3.
 - (D) 1.
12. Cross elasticity of a nearly perfect substitute products will be :
- (A) Infinite.
 - (B) Zero.
 - (C) > 1 .
 - (D) < 1 .
13. Cross elasticity of complementary products will be :
- (A) Infinite.
 - (B) Zero.
 - (C) > 1 .
 - (D) < 0 .

14. If the disposal income of a household decreases by 10 % and the demand for X commodity remains same. The income elasticity of X is :
- (A) 0. (B) 0.5.
(C) 0.5 (D) 2.5.
15. Which of these would lead to increase in quantity supplied at a given price :
- (A) Increase in VAT. (B) Increase in excise duty.
(C) Increase in import duty. (D) Reduction in levies.

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Mathematics

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(2019—2020 Admissions)

Time : Two Hours

Maximum : 60 Marks

Section A

Answer any number of questions.

2 marks each.

Maximum marks : 20.

1. What is disposable income ?
2. Write any two important determinants of demand.
3. Define elasticity of supply.
4. Point out the difference between a short run cost function and a long run cost function.
5. Write the relation between AR, MR and elasticity of demand.
6. Define average and marginal cost function.
7. Define utility function.
8. What is an indifference curve ?
9. Write any two criticisms against utility approach.
10. What is the major difference between the average concept and marginal concept in economics ?
11. Define (i) Increasing function ; and (ii) Decreasing function.
12. If $z = x^2 + 5xy$ find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.

Turn over

Section B

Answer any number of questions.

5 marks each.

Maximum marks : 30.

13. Explain arc price elasticity and point price elasticity.
14. With the help of a diagram explain the concept of supply.
15. Find out the output at which the average cost function is minimum from the total cost function $TC = 2q^2 + 5q + 18$.
16. Find the marginal and average functions of the total function given by $3Q^2 + 7Q + 12$ at $Q = 3$ and $Q = 5$.
17. Distinguish between cardinal and ordinal utility analysis.
18. Write the properties of indifference curve.
19. $f(x, y) = x^y$, verify that $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$.

Section C

*Answer any **one** question.*

The question carries 10 marks.

20. Define a demand function and explain all the variables in the demand function.
21. A firm producing two goods x and y has the profit function $\pi = 64x - 2x^2 + 4xy - 4y^2 + 32y - 14$. Find out the profit maximizing level of output for each of the two goods and test to be sure the profits are maximized.

(1 × 10 = 10 marks)

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MEC 1C 01—MATHEMATICAL ECONOMICS

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1. $y = 100/x + 4x$ has :
- (A) A maximum point where $x = 5$. (B) A minimum point where $x = 5$.
(C) A maximum point where $x = 0$. (D) A minimum point where $x = -5$.
2. Which of the following statements are (in general) true ?
- (A) Marginal Cost (MC) is minimised where $MC = \text{Average Variable Cost (AVC)}$.
(B) Total Cost (ATC) is minimised where $MC = ATC$.
(C) Average Variable Cost (AVC) is minimised where $MC = AVC$.
(D) Total revenue is maximised where $MC = \text{Marginal Revenue (MR)}$.
3. The law which studies the direct relationship between price and quantity supplied of a commodity is :
- (A) Law of demand. (B) Law of variable proportion.
(C) Law of supply. (D) None of the above.
4. When a percentage in price results in equal change in quantity supplied, it is called ?
- (A) Elastic supply. (B) Perfectly inelastic.
(C) Elasticity of supply. (D) Unitary elastic supply.
5. Few sellers is the feature of :
- (A) Monopoly. (B) Oligopoly.
(C) Perfect competition. (D) Monopolistic competition.
6. Supply curve of a perfectly competitive firm is :
- (A) Vertical. (B) Upward sloping.
(C) Horizontal. (D) Downward sloping.

7. Suppose the supply for product A is perfectly elastic. If the demand for this product increases:
- (A) The equilibrium price and quantity will increase.
 - (B) The equilibrium price and quantity will decrease.
 - (C) The equilibrium quantity will increase but the price will not change.
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8. If the demand curve for product A moves to the right, and the price of product B decreases, it can be concluded that :
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 - (C) (A) is an inferior good, and (B) is a superior good.
 - (D) Both goods (A) and (B) are inferior.
9. If a price increase of 50% results in an increase in the quantity supplied of an economic good from 10 to 20 pieces, calculate the co-efficient of price elasticity of supply.
- (A) 1/4.
 - (B) 1/2.
 - (C) 1.
 - (D) 2.
10. An economic agent contracts a loan of 15.000 lei, which he will repay in three equal annual installments. What will be the total interest paid, knowing that the annual interest rate is 12% per year ?
- (A) 3.600 lei.
 - (B) 1.800 lei.
 - (C) 5.400 lei.
 - (D) 1.500 lei.
11. Calculate the average fixed cost (AFC), for a level of production $Q = 20$, knowing that the total cost function is: $TC = 200 + 3Q + 2Q^2$.
- (A) 1060.
 - (B) 200.
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12. On the market with perfect competition :

- (A) The firm is a “price-taker”, meaning, it takes over the market price.
- (B) The firm is a “price-maker”, meaning, it determines the market price.
- (C) The companies’ products are differentiated.
- (D) Input barriers are minimal, and exit barriers are maximal.

13. Which of the following statements about monopoly is true :

- (A) There are several companies producing a specific product.
- (B) There is only one producing company, but the product has close substitutes.
- (C) There are no competitors on the relevant market.
- (D) Input barriers are low.

14. Which of the following can be considered as the basic features of public goods ?

- (A) Are state-owned;
- (B) Are characterized by non-excludability and non-rivalry;
- (C) Are characterized by excludability and rivalry;
- (D) May be positive or negative.

15. Which of the following solutions are not part of the ways of internalizing externalities ?

- (A) The imposition of fines on the producer of negative externalities.
- (B) The introduction of taxes and duties that bring private costs to the level of social costs.
- (C) Closure of companies producing positive or negative externalities.
- (D) The association of the negative externality manufacturer with the receptor of such an effect.

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Mathematics

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(2021 Admissions)

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Section A*Answer atleast eight questions.**Each question carries 3 marks.**All questions can be attended.**Overall ceiling 24.*

1. What is a demand curve ?
2. Define market demand curve.
3. What is meant by market equilibrium ?
4. Define cost elasticity.
5. Define average and marginal revenue.
6. Define elasticity of average cost.
7. What is a cardinal utility function ?
8. Define an indifference map.
9. Write any two properties of indifference curve.
10. Define total revenue, marginal revenue and average revenue.
11. Define : (i) A convex function ; (ii) A concave function.
12. If $f(x, y) = 3x^2y + x^3y^2$ find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.

(8 × 3 = 24 marks)

Turn over